CSE 4360/5364 Autonomous Robotics
Homework 1 – Fall 2026
Due 9/10/2026 11:59PM

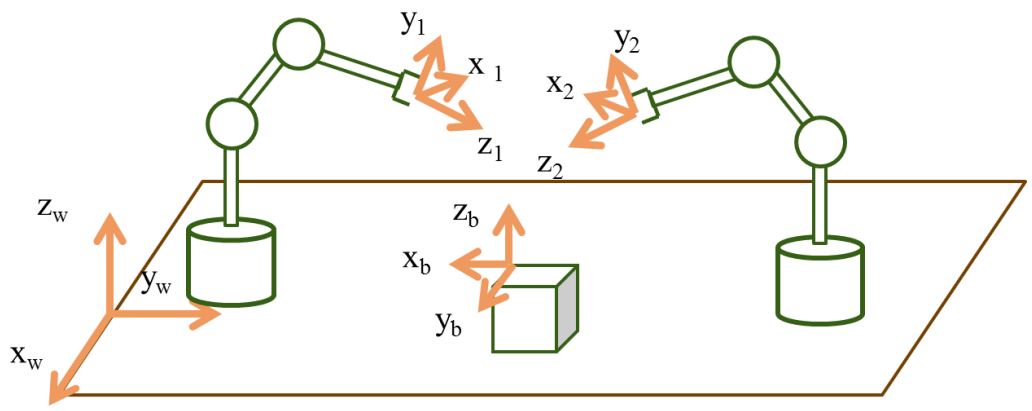
1) Verify equation 2.27 in the Textbook (2.26 in the 1st ed.))(4pts)

$$
\begin{aligned}
R_{ZYZ} &= R_{z,\phi}R_{y,\theta}R_{z,\psi} \\
&= \begin{bmatrix} c_\phi & -s_\phi & 0 \\ s_\phi & c_\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c_\theta & 0 & s_\theta \\ 0 & 1 & 0 \\ -s_\theta & 0 & c_\theta \end{bmatrix} \begin{bmatrix} c_\psi & -s_\psi & 0 \\ s_\psi & c_\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \\
&= \begin{bmatrix} c_\phi c_\theta c_\psi - s_\phi s_\psi & -c_\phi c_\theta s_\psi - s_\phi c_\psi & c_\phi s_\theta \\ s_\phi c_\theta c_\psi + c_\phi s_\psi & -s_\phi c_\theta s_\psi + c_\phi c_\psi & s_\phi s_\theta \\ -s_\theta c_\psi & s_\theta s_\psi & c_\theta \end{bmatrix} \qquad (2.27)
\end{aligned}
$$

2) Verify equation 239 in the Textbook (2.38 in the 1st ed.) (4pts)

$$
\begin{aligned}
R &= R_{z,\phi}R_{y,\theta}R_{z,\psi} \\
&= \begin{bmatrix} c_\phi & -s_\phi & 0 \\ s_\phi & c_\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c_\theta & 0 & s_\theta \\ 0 & 1 & 0 \\ -s_\theta & 0 & c_\theta \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & c_\psi & -s_\psi \\ 0 & s_\psi & c_\psi \end{bmatrix} \\
&= \begin{bmatrix} c_\phi c_\theta & -s_\phi c_\psi + c_\phi s_\theta s_\psi & s_\phi s_\psi + c_\phi s_\theta c_\psi \\ s_\phi c_\theta & c_\phi c_\psi + s_\phi s_\theta s_\psi & -c_\phi s_\psi + s_\phi s_\theta c_\psi \\ -s_\theta & c_\theta s_\psi & c_\theta c_\psi \end{bmatrix} \qquad (2.39)
\end{aligned}
$$

3)



Consider the figure above showing two robots engaged in a cooperative manipulation task. Suppose that

$$H_1^w = \begin{bmatrix} 0 & -1 & 0 & -13 \\ 1 & 0 & 0 & 14 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad H_2^w = \begin{bmatrix} 0 & -0.3827 & 0.9239 & -2 \\ 0 & 0.9239 & 0.3827 & -2 \\ -1 & 0 & 0 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

$$H_b^w = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Find H_1^2, H_b^1, and H_b^2. (4 pts each – total 12pts)

You may use Matlab, Python, etc., but try to do as much of it by hand as possible. Remember the rules for inverses of transforms are possible by hand.

Graduate Student Problems – All students in 5364 must do these two problem. Students in 4360 may do them for extra credit.

4) Show that the eigenvalues of a rotation matrix the x axis, R_x, are 1, $e^{\theta j}$, and $e^{-\theta j}$ where $j = \sqrt{-1}$ (5pts)

Hint – Solve the eigenvalues as usual. This is easy for the eigenvalue of λ=1. The other two are a bit harder. First, show that the remaining two roots are $\lambda = \cos(\theta) \pm j \sin(\theta)$. Then recall the identities

$$\sin x = \frac{e^{ix} - e^{-ix}}{2i}$$
$$\cos x = \frac{e^{ix} + e^{-ix}}{2}$$

The rest is easy.

It should be easy to see that this is true for R_z and R_y as well. As a matter of fact, all rotation matrices have eigenvalues 1, $e^{\theta j}$, and $e^{-\theta j}$, but this is harder to prove.

5) For a rotation matrix, what is the geometric interpretation of the eigenvector corresponding to the eigenvalue of 1? (2pts)

Hint – No calculations are necessary, just think about the definition of an eigenvector. Consider the specific case of R_x you solved in 4)